How to Flash a .hex File on an STM8S Minimum System Board

This comprehensive guide provides step-by-step instructions on how to flash a compiled **.hex** or **.ihx** firmware file onto an **STM8S103F3P6** minimum system development board (often misclassified as an ARM board) using an **ST-LINK V2 programmer** and **ST Visual Programmer (STVP)** software.

Required Tools and Equipment

- **Hardware:** STM8S103F3P6 "Blue Board" Minimum System Board
- **Hardware:** ST-LINK V2 Debugger/Programmer (or compatible clone)
- **Hardware:** 4 Female-to-Female Jumper Wires
- **Software:** ST Visual Programmer (STVP) with ST-LINK USB drivers
- **File:** Pre-compiled .hex or .ihx binary file (exported from your compiler or Arduino IDE)

Step 1: Hardware Connections

Disconnect your USB programmer from your PC before establishing connections. Connect 4 female-to-female jumper wires from the header pins on the ST-LINK V2 directly to the corresponding debug header pins along the edge of your STM8 blue board:

ST-LINK V2 Pin	STM8S Blue Board Pin	Description
SWIM	SWIM	Single Wire Interface Module Data line
RST / NRST	RST	Hardware Reset line
GND	GND	Common Ground reference
3.3V	3.3V	Microcontroller Power Delivery

WARNING: Never power the microcontroller board from multiple sources simultaneously. If you connect the 3.3V pin from the ST-LINK programmer to the board, do not plug an additional Micro-USB cable into the board's onboard USB port at the same time.

Step 2: Establish Software Configuration

1. Connect the ST-LINK V2 dongle to a USB port on your PC.
2. Launch the **ST Visual Programmer (STVP)** application.
3. If the Configuration prompt does not show automatically, navigate to **Configure** > **Configure ST Visual Programmer**.
4. In the **Hardware** column, scroll down and click **ST-LINK**.
5. In the **Port** column, choose **USB**.

6. In the **Programming Mode** column, choose **SWIM**.
7. In the **Device** list on the right side, select **STM8S103F3**.
8. Click **OK** to finalize settings.

Step 3: Load the HEX Binary File

1. Go to the top menu and select **File** > **Open**.
2. Browse to your project directory (or the build folder from your Arduino sketch export).
3. Choose your target .hex or .ihx file and click **Open**.
4. The workspace will update, showcasing rows of raw hexadecimal memory blocks populated inside the main window under the **PROGRAM MEMORY** tab.

Step 4: Flash the Microcontroller Memory

1. Double-check all physical wiring to guarantee communication will not break mid-transfer.
2. Select **Program** > **Current Tab** from the top menu, or click the **Program Current Tab** shortkey icon in the center toolbar (icon depicts an IC chip with a single red arrow pointing down).
3. The utility will automatically erase the target sector, run a blank check, and push the data frames.
4. Observe the console logs at the bottom. The flashing process is verified when you see:
   < PROGRAM MEMORY successfully verified.

Step 5: Power-Cycle to Run the Application

By design, STVP keeps control over the hardware reset line to lock the core from running code while connected to a programming instance. To unlock the chip:

1. **Close the ST Visual Programmer application completely.**
2. Disconnect the ST-LINK V2 programmer completely from the computer's USB port.
3. Wait 3 seconds for all internal caps to completely discharge.
4. Re-plug the ST-LINK back into the computer's USB port (or connect a clean Micro-USB power cable instead).
5. The microcontroller will instantly initialize and start running your firmware (e.g., blinking your onboard LED).